

AI-Assisted Development — Exam Submission

HotFixAmbulance — AI-Powered Production-Error Triage for .NET Web APIs

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AI tool used: Claude Code — Anthropic’s agentic CLI — running the **Claude Opus 4.8** model (1M-context)

Repository: <https://github.com/myPOSTech/mps-banking-hot-fix-ambulance>

Working-system evidence: <docs/evidence/working-system-evidence.pdf>

1. Project Overview

The idea

On-call engineers waste the first, most expensive minutes of an incident doing the same manual ritual: open Kibana, filter the last few hours of errors, eyeball the stack traces, guess the root cause, and go hunting through Git for the offending change. **HotFixAmbulance** automates that ritual. It is a triage tool that, for a chosen .NET Web API and time window, pulls the recent error logs, groups them into distinct problems, and — for each problem — uses a **local Qwen LLM** to write two human-readable columns: “**Suggestion for Error**” (what the error means) and “**How to fix**” (a concrete remediation), grounded in the service’s own Git history. The result is a ranked, explained, actionable error table in a web UI, plus a CLI for terminal/CI use.

System requirements

Functional

- Fetch Fatal/Error/Warning logs for a named API over a relative (last N h) or absolute time window from **Elasticsearch**.
- Group logs by exception fingerprint (exception type + normalized message + endpoint) and rank by severity, count, recency.
- For each group, produce an AI **Suggestion** and **How-to-fix**, using a **locally hosted Qwen model** and grounding the answer in Git blame + related commits.
- Persist every run; expose it through a REST API and a paginated React table; provide a CLI that triages and prints JSON/table output.
- The UI must make it **visible and verifiable** that a group was analysed by the LLM (a per-group 🤖 **Qwen** badge).

Non-functional

- **Graceful degradation:** the LLM is optional — if the model is unreachable, the system silently falls back to a deterministic Git-history heuristic and never fails a run.
- **Local & private:** the model runs on the developer’s machine in Docker (CPU-only), no data leaves the host.
- **Reproducible:** one command (`scripts/demo.ps1`) brings up the whole stack (Qwen, Elasticsearch, SQL Server), generates sample errors, runs triage, and opens the UI.

Tech stack

.NET 9 / C# 13 (backend, CLI) · ASP.NET Core Minimal APIs · Entity Framework Core + SQLite · Elasticsearch 8 ·

LibGit2Sharp · React 18 + TypeScript + Vite + TanStack Table + Tailwind · Ollama-runtime serving **Qwen 2.5 (3B)** ·

Docker Compose · PowerShell automation · xUnit + FluentAssertions + NSubstitute (backend) · Vitest + Testing Library (frontend).

2. System Architecture — Modules

The system was decomposed (with AI assistance) into focused technological modules, each a separate .NET project or front-end/infra unit with a single responsibility and a clear seam (interface) to its neighbours.

Testing strategy. Pure unit tests assert grouping keys, ranking order and correlation-id counting — fast and exhaustive because the layer is side-effect-free.

AI tool choice. Claude Code — strong at generating table-driven ranking tests and edge cases.

Prompts: “Group logs by a stable fingerprint and rank Fatal>Error>Warning, then count desc, then last-seen desc.” · “Add a normalizer that replaces GUIDs/numbers so the same error collapses into one group.”

3.3 AI Enrichment with Qwen (the centrepiece)

Approach & reasoning. The AI columns are owned by an `IGroupEnricher` seam so the strategy is swappable from config (`Analysis:Strategy`). `LlmGroupEnricher` builds a strict-JSON prompt (`{suggestion, howToFix}`) from the group + Git evidence, calls the model via `ILlmClient` (`OllamaLlmClient` , `temperature 0.2` , `format:"json"`), and on **any** failure (timeout, unreachable, bad JSON) degrades to the deterministic Git heuristic. A successful call tags the group `AnalyzedBy = "Llm"` ; that single fact drives the UI badge.

Step-by-step workflow. (1) `LlmPromptBuilder` (system contract + fact dump); (2) `OllamaLlmClient` that never throws (returns `null` on failure); (3) `LlmGroupEnricher` with fallback; (4) propagate per-group `AnalyzedBy` through `TriageService` → `ErrorGroup` → API → UI.

Testing strategy. Unit tests with a substituted `ILlmClient` assert: valid JSON → `Llm` ; null/garbage → heuristic fallback; and that `TriageService` stamps each group with the enricher’s source.

AI tool choice. Claude Code — it implemented the fallback contract, the run-level `Mixed/Llm/Heuristic` aggregation, and the tests in one TDD loop.

Prompts: “Add an LLM enricher that asks the model for {suggestion, howToFix} as JSON and falls back to the Git heuristic on any error — never throw.” · “Make sure every group records which strategy produced it, and surface a per-group marker in the UI proving Qwen was used.”

3.4 Git-History Grounding

Approach & reasoning. An LLM answer is only useful if grounded in the real code, so `FixHintBuilder` (behind `IFixHintSource`) uses `LibGit2Sharp` to fetch the offending file’s blame + a code snippet and search `origin/main` for related commits; this evidence is fed to both the heuristic and the Qwen prompt.

Step-by-step workflow. (1) `IGitRepoCache` to clone/pull per API; (2) `IGitHistoryReader` for blame + keyword commit search; (3) `FixHintBuilder` to assemble a compact evidence string.

Testing strategy. Unit tests substitute the repo cache/reader and assert the formatted hint contains the expected `file:line` , blame and commit SHAs.

AI tool choice. Claude Code — handled the `LibGit2Sharp` specifics and kept the evidence format testable.

Prompts: “Given a stack file+line, return blame + a code snippet from origin/main and the related commits.”

3.5 Persistence & REST API

Approach & reasoning. Runs are stored as a row with the groups serialised to JSON (SQLite), keeping the schema trivial and migration-free; the API is ASP.NET Core Minimal APIs with server-side paging/sorting (`GroupPager`).

Step-by-step workflow. (1) `HotFixDbContext` + `TriageRun` (incl. nullable `AnalyzedBy` for back-compat); (2) repository; (3) endpoints: run triage, latest, by-id, paged groups, history.

Testing strategy. Integration tests with `WebApplicationFactory` hit the real endpoints (in-memory/SQLite), asserting status codes, paging bounds and the `analyzedBy` field round-trips.

AI tool choice. Claude Code — generated endpoints + integration tests and kept DTOs in sync with the frontend types.

Prompts: “Expose paginated groups with sort/dir validation and an allowed page-size list, plus a run-by-id endpoint.”

3.6 Frontend UI + the Qwen badge

Approach & reasoning. A `TanStack-Table` grid with toggleable columns; the AI columns render a small violet

 Qwen badge iff `analyzedBy == 'Llm'` — the visible proof the model was used.

Step-by-step workflow. (1) typed API client + `ErrorGroup` type incl. `analyzedBy`; (2) `QwenBadge` component; (3) render it in the Suggestion and How-to-fix cells; (4) metrics panel (“AI Insights Generated”).

Testing strategy. Vitest + Testing Library: badge appears for `Llm`, is absent for `Heuristic` / `null` — written **before** the component (TDD red→green).

AI tool choice. Claude Code — wrote the failing tests first, then the component, matching the repo’s TDD rule.

Prompts: “Render a Qwen badge on the AI columns only when analyzedBy is 'Llm'; add Vitest tests for all three cases first.”

3.7 Infrastructure & DevOps

Approach & reasoning. Mirror the existing `infra/<service>` pattern: a CPU-only Qwen runtime in Docker Compose (`infra/qwen`), an idempotent `bootstrap.ps1` that pre-pulls `qwen2.5:3b`, and a `demo.ps1` that orchestrates the whole stack and **asserts** the produced run was analysed by the LLM before opening the UI.

Step-by-step workflow. (1) compose + healthcheck; (2) bootstrap (pull + chat probe); (3) corporate-CA injection (`export-host-ca.ps1` + `docker-compose.corp.yml`) for TLS-intercepting proxies; (4) wire `-WithLlm` /default LLM env into `demo.ps1`.

Testing strategy. Each script is verified live: compose `config`, container health, `/api/tags`, model present, idempotent re-run, and an end-to-end demo asserting `analyzedBy ∈ {Llm, Mixed}`.

AI tool choice. Claude Code — it ran the Docker/PowerShell commands, diagnosed failures from their output, and iterated.

Prompts: “Make Qwen part of the infrastructure in Docker like Elasticsearch, and guarantee the model is present before the app starts.” · “The container can’t verify TLS through our proxy — inject the corporate CA.”

3.8 Testing & Quality (cross-cutting)

Approach & reasoning. A mandatory TDD cycle (failing test first) and a **pre-commit hook** that blocks any commit unless `dotnet test`, `dotnet format`, frontend tests and lint all pass — so `main` /integration stays green.

Step-by-step workflow. Write the failing test → minimal implementation → run → commit (hook gates it).

Testing strategy. 160 backend unit tests + 18 integration tests + 34 frontend tests, all green at each commit.

AI tool choice. Claude Code — drove the red→green→commit loop and respected the project’s skills/hooks.

Prompts: “Add the failing test for AnalyzedBy first, watch it fail, then implement and run the suite.”

4. Challenges & Tool Comparison

Biggest challenges

- **Corporate TLS interception.** The Docker container could not `ollama pull` the model — `x509: certificate signed by unknown authority` — because the corporate proxy MITM-terminates TLS and the container lacked the corporate root CA. **Solved** by exporting the host CA bundle (`export-host-ca.ps1`) and mounting it via a `docker-compose.corp.yml` overlay with `SSL_CERT_FILE`. This was the single hardest, most environment-specific problem.
- **A latent DI bug surfaced by the new feature.** The CLI had never been wired to the `IGroupEnricher` seam added in a prior phase, so it crashed with “Unable to resolve service for type IGroupEnricher.” Found it by running the CLI; fixed by registering `AddHotFixGroupEnrichment` — which also enabled the CLI to use Qwen.
- **CPU inference latency.** One LLM call per group, sequential, on CPU (~10 s each) makes a 38-group run take minutes; mitigated by warming the model and raising the API client timeout. A real trade-off of running a private model locally.
- **Tooling papercuts.** PowerShell 5.1 mis-parsing a UTF-8 script because of em-dash characters (replaced with ASCII); Chrome headless needing absolute paths and an origin before `localStorage`; SQLite file-locks when a running API blocked the build. Each was diagnosed from command output and fixed.

Which tool helped the most, and why

Claude Code running the **Claude Opus 4.8** model (1M-context) was the primary and most valuable tool. Beyond code generation it could **run** the project — execute

Docker/PowerShell/dotnet, read the failures, and iterate — which is what actually solved the infrastructure problems

(the CA injection, the DI crash, the parse bug). Its skills/sub-agents kept the work disciplined (plan → TDD → verify →

commit with a passing pre-commit gate), and it could reason across the whole repo (backend + frontend + infra) to keep

types, DTOs and the demo orchestration consistent end-to-end.

What I would improve next

- **Batch / parallelise** the per-group LLM calls (or stream) to cut run time; optionally cache by fingerprint.
- Add a **GPU compose overlay** for much faster inference where a GPU is available.
- Make the persisted strategy tag itself read **Qwen** (currently the internal tag stays the generic **Llm**, while the UI shows “Qwen”).
- Share one SQLite store between CLI and API so a CLI-produced run is viewable in the UI without a second pass.
- Add a small **eval harness** that scores the model’s suggestions against known root causes.

5. Working System Evidence

See [docs/evidence/working-system-evidence.pdf](#) (4 captioned figures). Highlights:

- **Figure 1 — Overview:** a real run for **demo-api** — **73 logs** → **38 groups**, with **38 “AI Insights Generated”** and **38 “Fix Recommendations.”**
- **Figure 2 — Proof of Qwen:** every AI cell shows the  **Qwen** badge and genuine, context-specific model text

(e.g. “...failed database save operation at line 311 in DemoDatabase.cs...”, “...invalid SQL object name 'Pricing.Records'...”).

- **Figure 3 — Full table:** all **38 groups** carry the badge (run-level **analyzedBy = "Llm"**).
- **Figure 4 — Terminal:** **docker ps** (Qwen healthy on :11434), **ollama list** (**qwen2.5:3b** , 1.9 GB), API run

analyzedBy = Llm with **withSuggestions = 38** / **withFixes = 38** , and the container at **1177 % CPU** during inference.

Reproduce locally:

```
powershell -File scripts/demo.ps1 -WithElastic -KeepRunning
# then open the printed UI URL – every analysed group shows the  Qwen badge
```

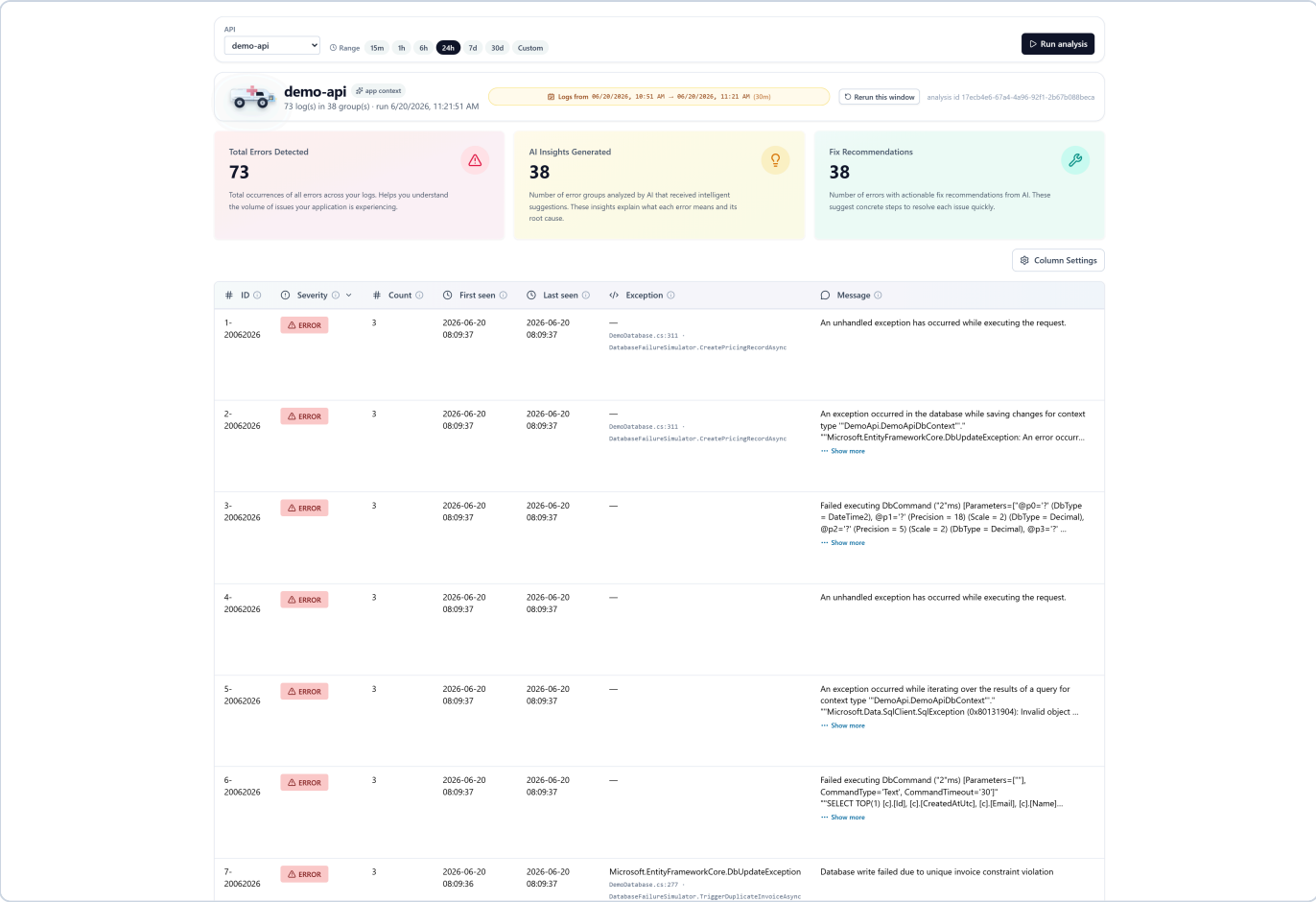


Figure 1 — System overview (ingestion, grouping, AI layer ran). A real triage run for the **demo-api** service: **73 error logs** collapsed into **38 distinct groups**. The metrics panel confirms the AI layer ran end-to-end — **38 "AI Insights Generated"** and **38 "Fix Recommendations"** (every group enriched). The table shows each group's severity, count, first/last-seen, exception with stack frame, message and endpoint — the raw evidence read from Elasticsearch. *Proves the ingestion → grouping → analysis pipeline works.*

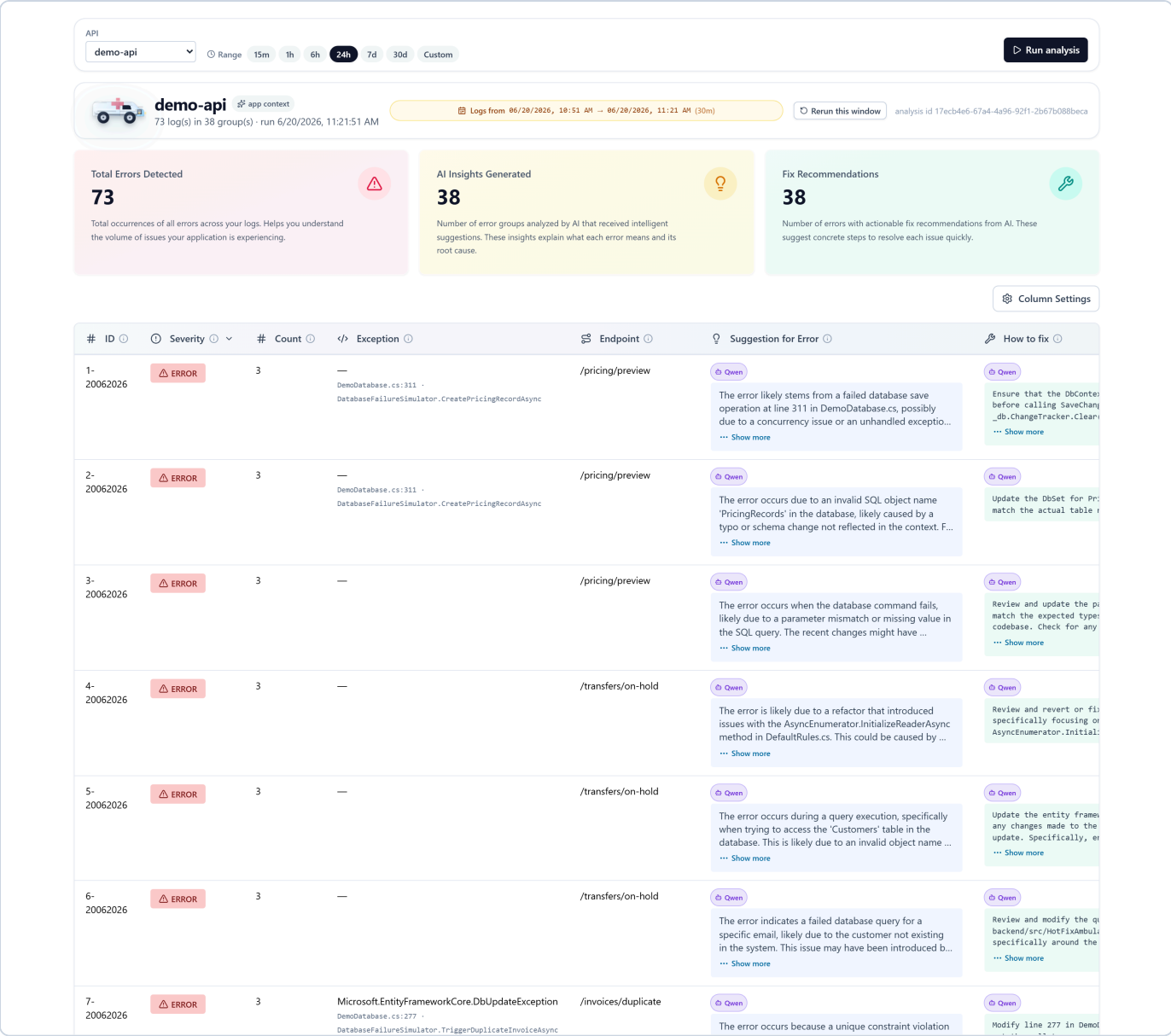


Figure 2 — Proof the analysis is produced by Qwen. The two right-most columns are the AI columns; each cell carries a violet  Qwen badge, rendered *only* when a group's analyzedBy == 'Llm' — a value the backend sets exclusively after a successful Qwen completion. The text is genuinely model-generated and context-specific (e.g. "...failed database save operation at line 311 in DemoDatabase.cs...", "...invalid SQL object name 'Pricing.Records'..."), not a static template. *Proves the system demonstrably uses Qwen.*

API

demo-api

Log In

Sign In

Get

Post

Put

Delete

Test

Docs

Admin

demo-api

API Explorer

71 logs in 30 groups

Run

Log In: 192.168.1.100, 14.10.40 ... 192.168.1.10, 11.11.41 (10s)

Run this whole

Snippet of Python 3.8.10 on 2024-03-26

Total Errors Detected

73

All Insights Generated

38

Fix Recommendations

38

Total accumulation of all errors across your logs. Help you understand the volume of issue your application is experiencing.

Number of error groups analyzed by AI that received intelligent suggestions. These logs to explain each error message and its root cause.

Number of errors with actionable fix recommendations for AI. These suggest concrete steps to resolve each issue quickly.

Columns Settings

#	ID	Severity	Count	Exception	Endpoint	Suggestion for Error	How to Fix
1	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error likely stems from a failed database save operation at line 311 in DemoDatabase.cs, possibly due to a concurrency issue or an unhandled exception.	Ensure that the ID exists before saving. Check the expected ID, add change detection, and use Show more
2	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs due to an invalid SQL object name 'Customers' in the database. Check the table name and the type of schema change not reflected in the context.	Update the object for the table. Verify the actual table. Show more
3	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs when the database command fails, likely due to a parameter mismatch or missing value in the SQL query. The recent changes might have ...	Review and update the query. Check the expected ID, add change detection, and use Show more
4	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error is likely due to a refactor that introduced issues with the ApplicationServices.InitializeBackend method in DemoDatabase.cs. This could be caused by ...	Review and refactor or fix specific logic. Review the ApplicationServices.InitializeBackend method. Show more
5	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs during a query execution, specifically when trying to access the 'Customers' table in the database. This is likely due to an invalid object name ...	Update the entity. From any changes made to the table, verify the table name. Show more
6	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error indicates a failed database query for a specific email. Likely due to the customer not existing in the system. This issue may have been introduced ...	Review and modify the database query. Verify the email address. Show more
7	2026/03/26	ERROR	3	Microsoft.EntityFrameworkCore.DbUpdateException: InvalidOperationException: (v: 27) IsNullOrEmpty() on 'Customer' (path=/api/customers/27)	/api/customers	The error occurs because a unique constraint violation is being thrown when trying to save changes to the database. Likely due to an existing record with the ...	Verify ID or 27 in ID. Verify the unique constraint. Show more
8	2026/03/26	ERROR	3	InvalidOperationException: (v: 27) IsNullOrEmpty() on 'Customer' (path=/api/customers/27)	/api/customers	The error occurs because a duplicate numeric member is being inserted into the 'Id' field, violating the unique constraint. This likely happened due to ...	Verify the value of 'id' in the database. Verify the unique constraint. Show more
9	2026/03/26	ERROR	3	Microsoft.Data.SqlClient.SqlException: The error occurs when the system fails to find a payment record by ID '40'. This likely stems from changes in the backend code, specifically around line ...	/api/payments	The error occurs when the system fails to find a payment record by ID '40'. This likely stems from changes in the backend code, specifically around line ...	Review and update the query. Verify the ID. Show more
10	2026/03/26	ERROR	3	Microsoft.Data.SqlClient.SqlException: The error occurs during a query execution, specifically when trying to read from the 'Payments' table. This likely stems from changes made in the ...	/api/payments	The error occurs during a query execution, specifically when trying to read from the 'Payments' table. This likely stems from changes made in the ...	Review and update the query. Verify the table name. Show more
11	2026/03/26	ERROR	3	Microsoft.Data.SqlClient.SqlException: The error indicates a failed database query, likely due to the PaymentId parameter not matching any records in the 'Payments' table. This issue may have been ...	/api/payments	The error indicates a failed database query, likely due to the PaymentId parameter not matching any records in the 'Payments' table. This issue may have been ...	Verify the value of 'id' in the database. Verify the unique constraint. Show more
12	2026/03/26	ERROR	3	Microsoft.Data.SqlClient.SqlException: The error occurs when the API fails to fetch a payment by ID '40'. Likely due to the recent refactoring in Phase 10, which may have introduced issues.	/api/payments	The error occurs when the API fails to fetch a payment by ID '40'. Likely due to the recent refactoring in Phase 10, which may have introduced issues.	Review and update the query. Verify the ID. Show more
13	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error indicates an invalid object name 'Payments' in the database query for context type 'DemoDbContext'. This likely stems from ...	Update the entity. Verify the table name. Show more
14	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs when the API fails to execute a SQL query for retrieving a payment by ID '311'. This likely stems from changes made in the refactoring of the ...	Review and update the query. Verify the ID. Show more
15	2026/03/26	ERROR	3	Microsoft.Data.SqlClient.SqlException: The error indicates an invalid object name 'Customers' in the SQL query, likely caused by a change in database schema or a typo. The log history points to ...	/api/customers	The error indicates an invalid object name 'Customers' in the SQL query, likely caused by a change in database schema or a typo. The log history points to ...	Review and update the query. Verify the table name. Show more
16	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs during a database query for the 'Customers' table, indicating an invalid object name. This likely stems from changes made in the ...	Review and modify the query. Verify the table name. Show more
17	2026/03/26	ERROR	3	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs when trying to select a customer by their default email, likely due to the SQL query failing because there is no record matching the provided ...	Review and modify the query. Verify the table name. Show more
18	2026/03/26	ERROR	2	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error indicates a 502 Bad Gateway HTTP response for the /api/payments/311 endpoint, likely caused by issues in the backend logic. This issue may ...	Review and modify the query. Verify the table name. Show more
19	2026/03/26	ERROR	1	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error indicates a 500 Internal Server Error on the /api/customers/311 endpoint, likely caused by an issue with saving changes to the database in ...	Review and modify the query. Verify the table name. Show more
20	2026/03/26	ERROR	1	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error indicates a server-side HTTP 500 Internal Server Error on the /api/customers/311 endpoint, likely caused by issues in the ...	Review and modify the query. Verify the table name. Show more
21	2026/03/26	ERROR	1	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The API endpoint '/api/customers/311' responded with a 500 Internal Server Error. Likely due to issues in the backend's SerilogLoggerMiddleware or the introduced ...	Review and modify the query. Verify the table name. Show more
22	2026/03/26	ERROR	1	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error likely stems from attempting to fetch a user with an invalid ID (3). This could be due to improper validation or mapping of the input parameters in the ...	Review and modify the query. Verify the table name. Show more
23	2026/03/26	ERROR	1	System.ArgumentOutOfRangeException: The error occurs when trying to look up a user with an invalid ID (3). This likely stems from the recent refactoring in DemoDatabase, where 'DemoDatabase ...	/api/customers	The error occurs when trying to look up a user with an invalid ID (3). This likely stems from the recent refactoring in DemoDatabase, where 'DemoDatabase ...	Review and modify the query. Verify the table name. Show more
24	2026/03/26	ERROR	1	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error indicates a server-side HTTP 500 Internal Server Error on the /api/customers/311 endpoint, likely caused by issues in the ...	Review and modify the query. Verify the table name. Show more
25	2026/03/26	ERROR	1	InvalidOperationException: (v: 311) IsNullOrEmpty() on 'Customer' (path=/api/customers/311)	/api/customers	The error occurs because the database save operation is not executed, leading to a concurrency violation and a potential data inconsistency or loss.	Review the call to the database. Verify the table name. Show more

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Figure 4 — Backend / runtime evidence (terminal).


```
# Qwen is served from Docker (CPU-only) on :11434
$ docker ps --filter name=hfa-qwen
hfa-qwen    ollama/ollama:latest    Up 59 minutes (healthy)    0.0.0.0:11434→11434/tcp

# qwen2.5:3b is pulled into the persistent volume
$ docker exec hfa-qwen ollama list
NAME          ID          SIZE      MODIFIED
qwen2.5:3b    357c53fb659c  1.9 GB    37 minutes ago

# A triage run produced via the API – analysed by the LLM, every group enriched
$ curl -s http://localhost:5283/api/triage/runs/17ecb4e6-... /
apiName       = demo-api
analyzedBy    = Llm
totalLogs     = 73
totalGroups   = 38
summary.withSuggestions = 38      # all 38 groups got an AI suggestion
summary.withFixes      = 38      # all 38 groups got an AI fix

# During inference the Qwen container saturates the CPU (one call per group)
$ docker stats hfa-qwen --no-stream
hfa-qwen    CPU = 1177%    MEM = 4.4 GiB / 15.4 GiB
```

Proves Qwen runs locally in Docker, the model is loaded, and a real run returns `analyzedBy = Llm` with an AI suggestion + fix for every one of the 38 groups.

6. Repository

GitHub: <https://github.com/myPOSTech/mps-banking-hot-fix-ambulance>

(Feature work on branch `integration-llm`.)

7. Author's Notes

This is **my own idea project**. Beyond the exam, I intend to present it to **myPOS management** as a **working prototype** for everyday use by the **.NET teams** — a practical, privacy-respecting way to shorten incident triage in a fintech environment where production data must stay in-house.

Why it is worth adopting:

- **Faster incident response (lower MTTR).** On-call engineers open a ranked, *explained* error list instead of scrolling raw Kibana logs — the “what” and the “how to fix” are already written for each distinct problem.
- **Lower skill barrier.** Junior engineers get plain-English root-cause hints and remediation steps ***grounded in the team’s own Git history*** (blame + related commits), not generic advice.
- **Consistent, repeatable triage.** Every incident is analysed the same way, with the same evidence, so handovers and post-mortems start from a common baseline.
- **Privacy & compliance by design.** The model runs **locally in Docker (CPU-only)** — no logs, stack traces or source context leave myPOS infrastructure. Critical for banking/PCI contexts where cloud LLMs are a non-starter.
- **No per-token cost.** Uses a small open model (`qwen2.5:3b`) locally; there are no cloud-API usage fees.
- **Fits the existing stack.** .NET, Serilog, Elasticsearch and Git are tools the teams already run; adoption is additive, not a migration.
- **Safe to roll out.** The LLM is optional and **degrades gracefully** — if the model is down, triage still works via the deterministic Git heuristic, so it can be enabled per-team without risk.
- **Extensible.** Pluggable analysis strategy (heuristic ↔ LLM), pluggable model and per-API configuration mean it can grow from a prototype to a shared internal service.

Proposed next step at myPOS: pilot it with one .NET team on a non-critical service for two weeks, measure the

change in triage time and engineer feedback, then decide on a shared, GPU-backed internal deployment.

8. How to Reproduce the Full End-to-End Test

This section lets a reviewer reproduce the **entire experimental setup** shown in §5 from a clean checkout. One PowerShell command brings up the whole stack (Qwen + Elasticsearch + SQL Server), generates sample errors, runs the triage on Qwen, starts the API + UI, asserts the run was analysed by the LLM, and opens the browser on it.

8.1 Prerequisites

- **Windows 10/11 + PowerShell 5.1** (the automation uses Windows/PowerShell features).
- **Docker Desktop** installed and **running** (Linux containers).
- **.NET 10 SDK, Node.js 18+ & npm, Git**.
- **Disk/network:** the first run downloads ~**6 GB** of images + models (Ollama image, **qwen2.5:3b** ≈ 2 GB,

Elasticsearch, SQL Server). Subsequent runs reuse the cached volumes and are much faster.

- A free **11434** (Qwen), **5283** (API), **5173** (frontend), **9200** (Elasticsearch), **14333** (SQL Server).

8.2 One-time setup

```
# 1) Clone and switch to the feature branch
git clone https://github.com/myPOSTech/mps-banking-hot-fix-ambulance.git
cd mps-banking-hot-fix-ambulance
git checkout integration-llm

# 2) Restore tooling and install the pre-commit hook (.NET restore + npm install)
powershell -File scripts/bootstrap.ps1

# 3) Make sure Docker Desktop is running
docker version
```

8.3 Run the full end-to-end demo

```
powershell -File scripts/demo.ps1 -WithElastic -KeepRunning
```

This single command performs, in order:

1. **Starts the Qwen runtime** (**infra/qwen** , Docker, CPU-only) and runs its bootstrap, which **pulls qwen2.5:3b** if absent and verifies it with a JSON **/api/chat** probe. (*First run downloads ~2 GB — be patient*)
2. Sets **Analysis:Strategy=Llm** + the **Llm** endpoint/model for every child process.
3. **Starts SQL Server and Elasticsearch** (Docker) and waits for health.
4. **Builds and starts demo-api** , hammers its endpoints to **generate realistic error traffic**, then reshapes the logs into Elasticsearch via the ingest pipeline.
5. **Runs the CLI triage** on Qwen, then **builds and starts the API (:5283) and the frontend (:5173)**.
6. **Creates a triage run via the API** and **asserts** the result's **analyzedBy** is **Llm / Mixed** — i.e. it **fails loudly if Qwen was not actually used**.
7. **Opens your default browser** on the produced run (see §8.5 for the link and what is shown) — every analysed group shows the  **Qwen** badge.

8.4 Expected console output (key markers)

```
[demo] LLM strategy enabled: provider=Qwen model=qwen2.5:3b endpoint=http://localhost:11434
[qwen-bootstrap] chat probe: ok
[demo] Producing error traffic
[es-bootstrap] Reindex: total=94 created=94 updated=0 failures=0
[demo] Running CLI: hot-fix-ambulance demo-api --lookback 1h
[demo] Starting HotFixAmbulance.Api on http://localhost:5283
[demo] Creating the run via API: POST /api/triage/demo-api?lookbackHours=1
[demo] API run id=<guid> analyzedBy=Llm groups=38
[demo] Verified: this run was analyzed by Qwen (analyzedBy=Llm). The UI shows the Qwen badge.
[demo] Opening UI: http://localhost:5173/?analysisId=<guid>&api=demo-api
```

8.5 Open the result in the browser

When the demo finishes it **automatically opens your default browser** on the produced run. The URL looks like:

```
http://localhost:5173/?analysisId=<run-guid>&api=demo-api
```

The exact `<run-guid>` is printed in the console (see §8.4 — the `Opening UI:` line). While the API (`:5283`)

and the frontend (`:5173`) stay running you can also open the result manually at any time:

- **Latest run, no id needed:** open `http://localhost:5173/?api=demo-api` — the UI loads the most recent run for `demo-api`.
- **A specific run:** paste the full URL using the `analysisId` printed in the console.

What you will see (Figure 5). The triage dashboard for `demo-api`: a header with the run summary

(**73 log(s) in 38 group(s)**), a metrics strip (**Total Errors Detected**, **AI Insights Generated**,

Fix Recommendations), and the ranked error table. In the “**Suggestion for Error**” and “**How to fix**”

columns, **every analysed group shows a violet  Qwen badge** above the model-written text — the visible proof

that the Qwen LLM produced the analysis. The **Column Settings** button (top-right) toggles which columns are

shown; the **Run analysis** button (top-right) re-runs the triage for the selected API and time range.

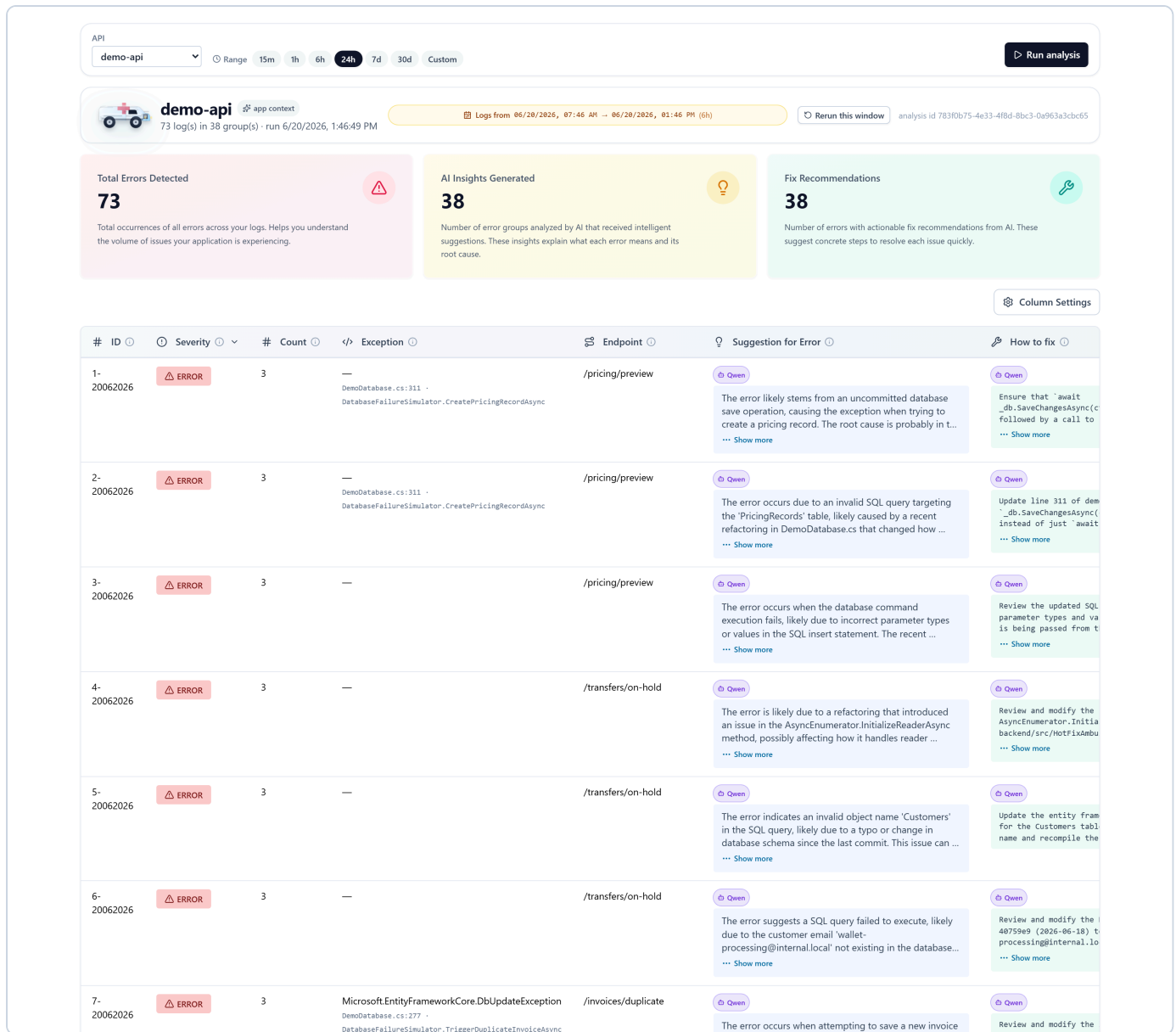


Figure 5 — The browser opened on the produced run: the demo-api triage dashboard. Every group's "Suggestion for Error" and "How to fix" cell shows a Qwen badge above the model-generated text, proving the analysis came from the Qwen LLM.

8.6 Verify the LLM directly (optional, independent proof)

```
# The Qwen model is loaded in the container
docker exec hfa-qwen ollama list          # → qwen2.5:3b ... 1.9 GB

# Produce a run via the API and confirm it was analysed by the LLM
curl.exe -s -X POST "http://localhost:5283/api/triage/demo-api?lookbackHours=1"
# → JSON whose "analyzedBy" field is "Llm"

# Watch the Qwen container saturate the CPU during inference
docker stats hfa-qwen --no-stream        # → CPU ~1000%+ while a run is in flight
```

8.7 Troubleshooting

- x509: certificate signed by unknown authority on model pull (corporate networks).** `demo.ps1` already handles this — it exports the host CA bundle (`infra/qwen/export-host-ca.ps1`) and starts the runtime with `infra/qwen/docker-compose.corp.yml` so the container trusts an intercepting proxy. On a normal home network no action is needed.
- docker compose up failed** . Docker Desktop is not running — start it and re-run.

- **A port is busy.** Stop whatever holds `5283` / `5173` / `11434` , or close a previous demo run.
- **Model pull is slow.** It is a one-time ~2 GB download; the demo waits for it. Re-runs skip it.
- **Run without the LLM** (heuristic only, no Docker model): `powershell -File scripts/demo.ps1 -WithElastic -KeepRunning -SkipLlm` .

8.8 Teardown

```
docker compose -f infra/qwen/docker-compose.yml down          # keeps the model volume
docker compose -f infra/elasticsearch/docker-compose.yml down
docker compose -f infra/mssql/docker-compose.yml down
# add '-v' to any of the above to also delete its data/model volume
```

9. Project Timeline (from the Git history)

The project was built in **69 commits over four days (16–20 June 2026)**. Every commit was gated by a pre-commit hook that runs the full backend + frontend test suites, formatting and lint — `--no-verify` was never used — so each row below is a green, independently-inspectable checkpoint. The history is generalised by day, with the milestone reached and a few representative commit ids.

Day	Milestone	Key commits	What was done
16 Jun	End-to-end skeleton working	<code>981a210</code> , <code>bb62f32</code> , <code>0cd65da</code> , <code>b213806</code> , <code>2b061dd</code> , <code>bc7b613</code> , <code>b773ed7</code>	Scaffolded the repo + Claude tooling, then built the backend solution and Core domain (Severity / LogEntry / ErrorGroup, 16 tests) and every module in turn — Elastic ingestion, heuristic grouping, Git-insights (LibGit2Sharp), persistence + Minimal API, and the CLI — plus the React triage table, the demo-api with Serilog, and a Dockerized Elasticsearch stack. First full pipeline: ingest → analyse → git evidence → API / UI / CLI.
18 Jun	Realistic demo data + time windows	<code>acd1c28</code> , <code>40759e9</code> , <code>0a3b622</code> , <code>66a932e</code> , <code>d54056f</code>	Rewrote demo-api for real EF Core CRUD against Dockerized SQL Server (genuine DB failures: unique-constraint, timeouts); surfaced the offending source line + git blame in “How to fix”; added the <code>TimeWindow</code> value object (relative/absolute ranges), the frontend <code>TimeRangePicker</code> + API dropdown, and expandable cells.
19 Jun	Server-side pagination + LLM groundwork	<code>c5dd730</code> , <code>13e7027</code> , <code>b923f5c</code> , <code>e5670e9</code> , <code>d0111a6</code>	Shipped server-side pagination (<code>PagedResult</code> / <code>TriageSummary</code> / <code>GroupPager</code> , paged groups endpoint, <code>TriageRunHeader</code> , numbered footer); fixed fractional lookback hours; scaffolded the <code>HotFixAmbulance.Llm</code> project and the JSON-contract prompt builder.
20 Jun	Whole flow on local Qwen + visible proof + docs	<code>3831bc8</code> , <code>9d7f17e</code> , <code>c15520b</code> , <code>979252b</code> , <code>7e25e9a</code> , <code>3657a19</code>	Completed the LLM layer (<code>ILlmClient</code> / <code>OllamaLlmClient</code> , the <code>IGroupEnricher</code> seam, <code>LlmGroupEnricher</code> with graceful fallback, the <code>Analysis:Strategy</code> toggle); added the per-group <code>AnalyzedBy</code> marker and the 🚑 badge, then rebranded it to Qwen ; fixed a latent CLI DI break; Dockerised the Qwen runtime with model bootstrap + corporate-CA injection; ran the whole demo on Qwen and proved it via the UI badge; and produced the full exam documentation (write-up, evidence, merged PDF).

Reviewer note. The history reads as a disciplined, test-first progression: a working vertical slice first (16 Jun), then realism and UX (18 Jun), then scale/pagination and LLM groundwork (19 Jun), and finally the AI feature delivered end-to-end with proof and documentation (20 Jun). Each phase is a small, independently-tested commit, so any single step can be reviewed in isolation, and the green pre-commit gate on every commit means the `main` /integration branch was never left broken.